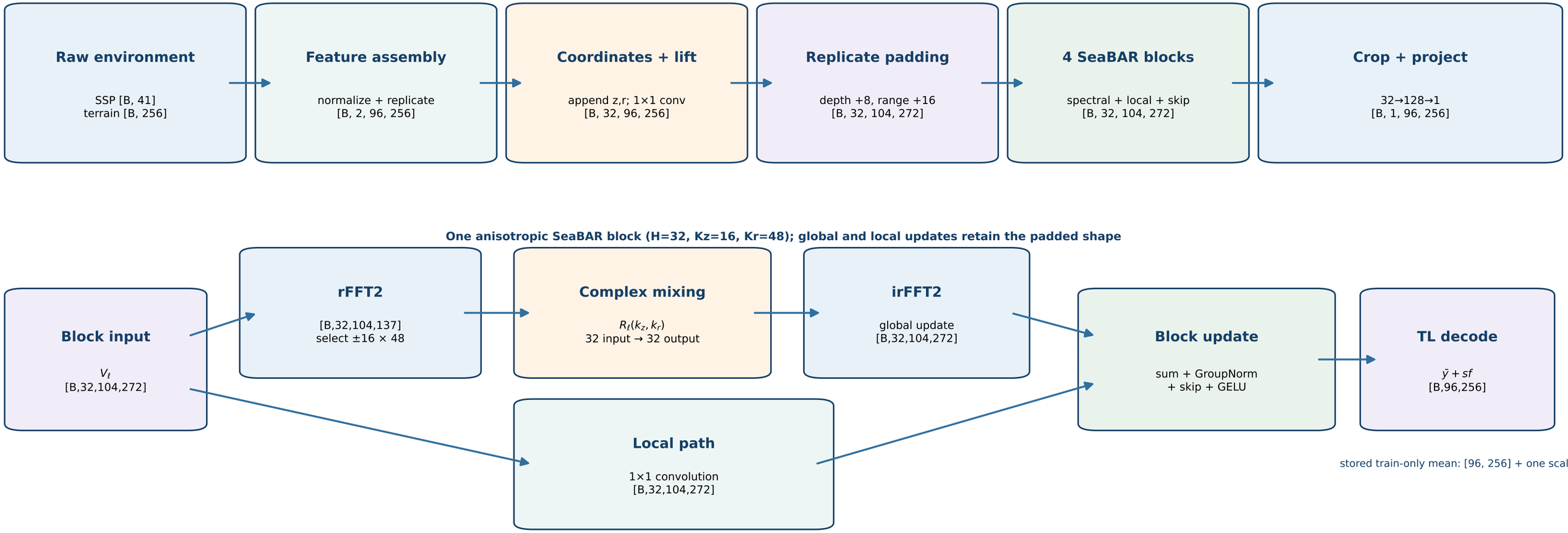
Raw environment
SSP [B, 41]
terrain [B, 256]
Feature assembly
normalize + replicate
[B, 2, 96, 256]
Coordinates + lift
append z,r; 1×1 conv
[B, 32, 96, 256]
Replicate padding
depth +8, range +16
[B, 32, 104, 272]
4 SeaBAR blocks
spectral + local + skip
[B, 32, 104, 272]
Crop + project
32→128→1
[B, 1, 96, 256]
One anisotropic SeaBAR block (H=32, Kz=16, Kr=48); global and local updates retain the padded shape
Block input
V_ℓ
[B,32,104,272]
rFFT2
[B,32,104,137]
select ±16 × 48
Complex mixing
$R_\ell(k_z, k_r)$
32 input → 32 output
irFFT2
global update
[B,32,104,272]
Local path
1×1 convolution
[B,32,104,272]
Block update
sum + GroupNorm
+ skip + GELU
TL decode
$\bar{y} + sf$
[B,96,256]
stored train-only mean: [96, 256] + one scale